

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE : MED-I (B)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their **Form No.** in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

Test Booklet Code

This Booklet contains **28** pages.

E1

Important Instructions :

Do not open this Test Booklet until you are asked to do so.

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - (a) **Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - (b) **Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.

Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
3. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720.**
4. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
5. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
6. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
7. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
8. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
9. Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
10. No candidate, without special permission of the Invigilator, would leave his/her seat.
11. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
12. Use of Electronic/Manual Calculator is prohibited.
13. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
14. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
15. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
16. Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____
: in words _____

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Candidate's Signature : _____ **Invigilator's Signature :** _____

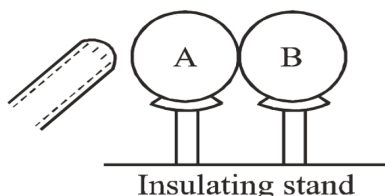
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SECTION-A (PHYSICS)

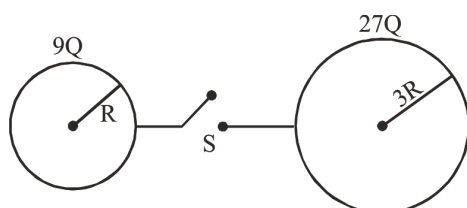
1. Electrostatic force and gravitational force differ in which respect?

- (1) Conservative force
- (2) Central force
- (3) Principle of superposition
- (4) Dependence on the intervening medium

2. Two uncharged metal spheres A and B are in contact as shown in the diagram. A negatively charged rod is brought near to A, but not touching it. The two spheres are separated slightly and the rod is then withdrawn. As a result of this :-

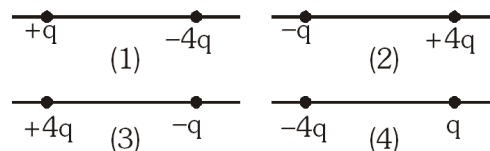


- (1) both the spheres acquire +ve charge
 - (2) both the spheres acquire -ve charge
 - (3) A acquires -ve and B acquires +ve charge
 - (4) A acquires +ve and B acquires -ve charge
3. If a body is charged by rubbing it, its weight :-
- (1) Increases slightly
 - (2) Decreases slightly
 - (3) Remain precisely constant
 - (4) May increase slightly or may decrease slightly
4. Find final charge on smaller sphere after closing switch :-



- (1) 18 Q (2) 27 Q (3) 9 Q (4) zero

5. The figure shows four situations in which charges as indicated ($q > 0$) are fixed on an axis. In which situation is there a point to the left of the charges where a charge would be in equilibrium ?

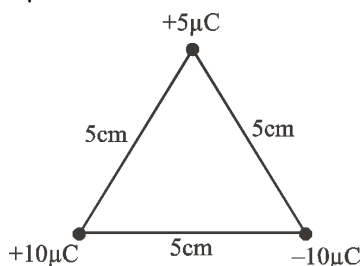


- (1) 1 and 2
 - (2) 2 and 4
 - (3) 3 and 4
 - (4) 1 and 3
6. There are two charges $+2nC$ and $+7nC$. The ratio of the forces acting on them will be :-
- (1) 1 : 1
 - (2) 2 : 7
 - (3) 7 : 2
 - (4) 4 : 49
7. Two point charges exert a force 20 N when placed in a free space. When this system is placed in a medium has dielectric constant 5, then they exert the force on each other is :
- (1) $20 \times \sqrt{5}$ N
 - (2) 4 N
 - (3) $\frac{20}{\sqrt{5}}$ N
 - (4) 5 N
8. Two point charges $3\mu C$ and $4\mu C$ repel with a force of 10 N. If $-6\mu C$ charge is given to each, find the new value of force between them :-
- (1) 2N
 - (2) 4N
 - (3) 5N
 - (4) 7.5N

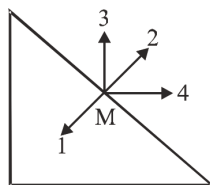
9. Dimensions of permittivity of free space (ϵ_0)

- (1) $[M^{-1}L^{-3}T^4A^2]$
- (2) $[ML^3T^{-4}A^{-2}]$
- (3) $[M^{-1}L^{-3}T^{-2}A^{-2}]$
- (4) None of these

10. In the given arrangement of charge, find net force on $+5\mu\text{C}$



- (1) 90 N (2) 180 N
(3) 0 N (4) $90\sqrt{3}$ N
11. Three identical positive point charges, as shown are placed at the vertices of an isosceles right angled triangle. Which of the numbered vectors coincides in direction with the electric field at the mid-point M of the hypotenuse:-

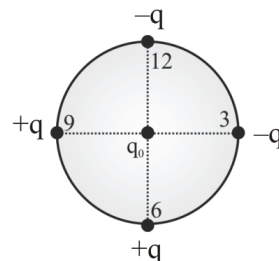


- (1) 1 (2) 2 (3) 3 (4) 4
12. Electrostatic force between two charges is 200 N in vacuum. When both charges are placed in medium at same distance the force becomes 80 N, ϵ_r will be equal to-
- (1) 1 (2) 1.25 (3) 2.5 (4) 0
13. Two charges of same magnitude but of opposite nature are placed by some distance away. F newton force is acting between them. If 50% of one charge is transferred to another charge. Then force on them will be :-

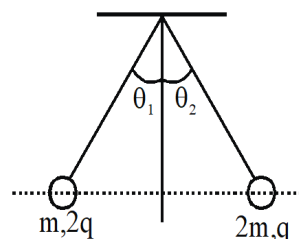
- (1) $\frac{3}{4}F$ (2) $\frac{4}{3}F$ (3) $4F$ (4) $\frac{F}{4}$
14. Two particles A and B having equal charges are placed at a distance $2m$ apart. A third charged particle placed on the perpendicular bisector at a distance x will experience the maximum Coulomb's force when :-

- (1) $x = 2m$ (2) $x = 1m$
(3) $x = \frac{1}{\sqrt{2}}m$ (4) $x = \sqrt{2}m$

15. Four charges are placed at the circumference of dial clock as shown in figure. If the clock has only hour hand, then the resultant force on a charge q_0 placed at the centre, points in the direction which show the time as :-

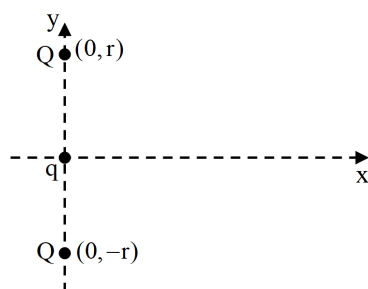


- (1) 1:30 (2) 4:30 (3) 7:30 (4) 10:30
16. Two particles which have masses and charges m and $2q$ and $2m$ and q respectively are hung by massless threads from a point as shown in equilibrium particles are on same horizontal line what can be the relation in θ_1 & θ_2 . (θ_1 & θ_2 are not up to scale in figure)



- (1) $\theta_1 > \theta_2$
(2) $\theta_1 < \theta_2$
(3) 0.18 N
(4) More information is required.
17. **Assertion (A) :-** Two similarly charged bodies may attract each other.
Reason (R) :- When charge on one body (Q) is much greater than that on another (q) and they are close enough to each other, then force of attraction between Q and induced charges exceeds the force of repulsion between Q and q .
- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(2) (A) is correct but (R) is not correct
(3) (A) is incorrect but (R) is correct
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

18.

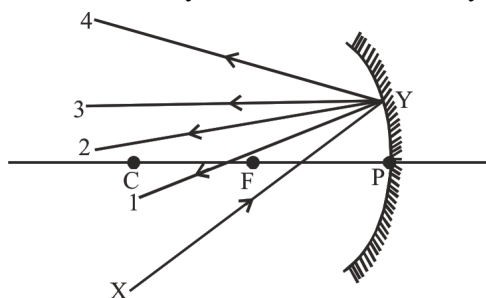


All charges are of similar type. when q charge displaced from origin. Then choose correct options.

- (1) It is in stable equilibrium along x-axis.
- (2) It is in unstable equilibrium along x-axis.
- (3) It is in stable equilibrium along y-axis.
- (4) It is in unstable equilibrium along y-axis.

(1) 2, 4 (2) 1, 3 (3) 2, 3 (4) 1, 4

19. Figure shows a small concave mirror with CP as its principal axis. A ray XY is incident on the mirror. Which of the four rays can be the reflected ray :-

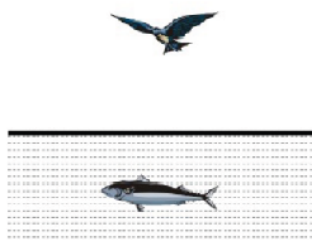


(1) 1 (2) 2 (3) 3 (4) 4

20. The time required to pass the light through a glass slab of 2 mm thick is ($\mu_{\text{glass}} = 1.5$)

- (1) 10^{-5} s (2) 10^{-11} s
- (3) 10^{-9} s (4) 10^{-13} s

21. A bird hovering over a lake sees a fish at a distance of 1.2 m from itself. The fish sees the bird at distance of 1.5 from itself. What is the refractive index of water ?



- (1) $\frac{5}{4}$ (2) $\frac{4}{3}$ (3) $\sqrt{\frac{5}{4}}$ (4) $\sqrt{\frac{4}{3}}$

22.

A container of depth H is filled with two immiscible transparent liquids of refractive index μ_1 and μ_2 respectively. The depth of each liquid is $H/2$. When viewed from above the apparent depth of the vessel is :-

- (1) $\frac{H}{2\mu_1} + \frac{H}{2\mu_2}$
- (2) $\frac{H}{2\mu_1} - \frac{H}{2\mu_2}$
- (3) $\frac{H}{2\mu_1} + \frac{H\mu_1}{2\mu_2}$
- (4) $\frac{H}{2\mu_1} - \frac{H\mu_1}{2\mu_2}$

23.

An object of size 3 cm is placed 14 cm in front of a convex mirror of focal length 21 cm. Size of the image when object is moved farther from the mirror is

- (1) Increasing
- (2) Decreasing
- (3) Same
- (4) First decreasing then increasing

24.

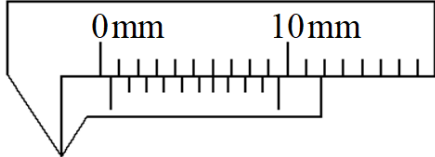
A particle is moving towards a fixed spherical mirror. The image:

- (1) must move away from the mirror
- (2) must move towards the mirror
- (3) may move towards the mirror
- (4) will move towards the mirror, only if the mirror is convex.

25.

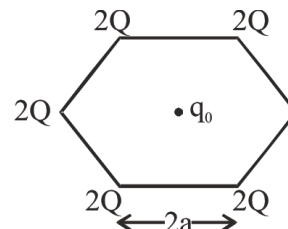
Sun subtends an angle of 0.5° at the pole of a concave mirror of radius of curvature 15 m. The diameter of the image of the sun formed by the mirror is :

- (1) 8.55 cm
- (2) 7.55 cm
- (3) 6.55 cm
- (4) 5.55 cm

26. A point source is placed 15 cm away from a convex mirror. A virtual image is formed at a distance of 6 cm. The radius of curvature of the mirror is-
- (1) 4.3 cm (2) 8.6 cm
(3) 10 cm (4) 20 cm
27. If refractive index of a medium is 1.5, then velocity of light in that medium will be :
- (1) 10×10^8 (2) 2×10^8
(3) 3×10^8 (4) 4×10^8
28. The wavelength of light in two liquids 'x' and 'y' is $3500 \text{ \AA}$ and $7000 \text{ \AA}$, then the critical angle will be
- (1) 60° (2) 45°
(3) 30° (4) 15°
29. A microscope is focused on a mark, then a glass slab of refractive index 1.5 and thickness of 6 cm is placed on the mark to get the mark again in focus, the microscope should be moved
- (1) 4 cm (2) 2 cm
(3) 6 cm (4) 8 cm
30. The frequency of a light wave in a material is $4 \times 10^{14} \text{ Hz}$ and wavelength is $5000 \text{ \AA}$. The refractive index of material will be :
- (1) 1.33 (2) 1.40
(3) 1.50 (4) 3.00
31. The external and internal radius of a hollow cylinder are measured to be $(8.00 \pm 0.08) \text{ cm}$ and $(3.00 \pm 0.01) \text{ cm}$. The thickness of the wall of the cylinder is :-
- (1) $(11.00 \pm 0.09) \text{ cm}$
(2) $(11.00 \pm 0.07) \text{ cm}$
(3) $(5.00 \pm 0.07) \text{ cm}$
(4) $(5.00 \pm 0.09) \text{ cm}$
32. A screwgauge has a least count of 0.02 mm and number of circular scale divisions on its circular scale are 100. Find pitch of screw of screwgauge:
- (1) 1 mm (2) 2 mm
(3) 0.5 mm (4) 4 mm
33. The resistance R of a wire is calculated using the formula $R = \frac{V}{I}$ where $V = (100 \pm 5) \text{ volt}$ and $I = (20 \pm 0.4) \text{ Ampere}$. The Resistance R will be (approximately) :-
- (1) $(5 \pm 0.14) \Omega$ (2) $(5 \pm 2.5) \Omega$
(3) $(5 \pm 0.4) \Omega$ (4) $(5 \pm 0.7) \Omega$
34. Find the zero error in the vernier calliper, if main scale has mm marks and 10 divisions of vernier scale coincide with 9 divisions of main scale :-
- 
- (1) 0.4 mm (2) 0.6 mm
(3) -0.4 mm (4) -0.6 mm
35. Least count error belongs to category of -
- (1) Random error (2) Systematic error
(3) Both (1) and (2) (4) None of these

SECTION-B (PHYSICS)

36. Find magnitude of the force on a point charge q_0 placed at the centre of regular Hexagon of side $2a$.



- (1) $\frac{2KQq_0}{a^2}$ (2) $\frac{KQq_0}{a^2}$
(3) $\frac{KQq_0}{2a^2}$ (4) $\frac{KQq_0}{4a^2}$

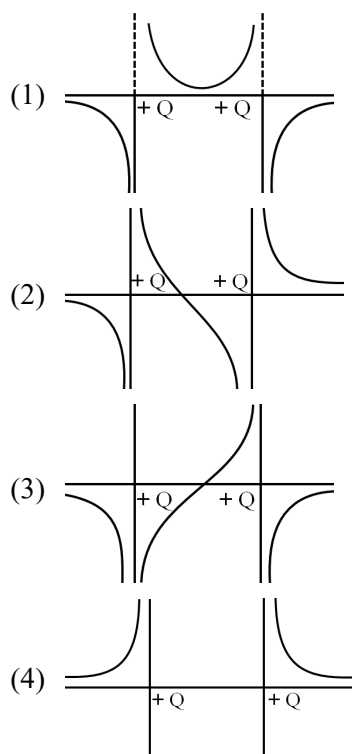
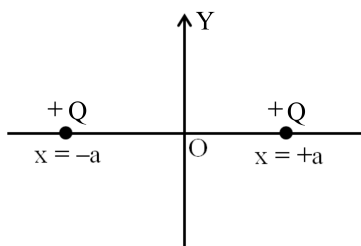
37. Two identical charged spheres are suspended by strings of equal length. The strings make an angle of 30° with each other. When suspended in a liquid of density 0.8 g/cc , the angle remains the same. What is the dielectric constant of the liquid ? [Density of the material of sphere is 1.6 g/cc .] :-

(1) 1.5 (2) 2 (3) 3 (4) None

38. A point charge of $25 \mu\text{C}$ is located in the XY plane at the point of position vector $\vec{r}_0 = (\hat{i} + \hat{j}) \text{ m}$. What is the magnitude of electric field at the point of position vector $\vec{r}_1 = (4\hat{i} + 5\hat{j}) \text{ m}$?

(1) $900 \frac{\text{V}}{\text{m}}$ (2) $9 \frac{\text{V}}{\text{m}}$
(3) $90 \frac{\text{V}}{\text{m}}$ (4) $9000 \frac{\text{V}}{\text{m}}$

39. Two identical positive point charge placed on x-axis the variation of E along x-axis.



40. In the given diagrams the direction of electric field at point O is given in list-II (O is circumcenter of the given regular polygon). Charge Q is positive. Match the direction of electric field for the given arrangement:-

List-I		List-II	
(P)		(1)	
(Q)		(2)	
(R)		(3)	
(S)		(4)	

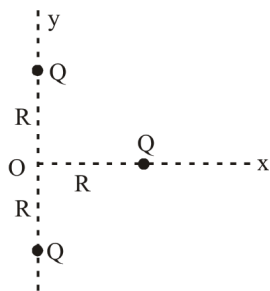
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	P	Q	R	S
(1)	2	1	4	3
(2)	3	1	2	4
(3)	3	2	1	4
(4)	1	4	3	2

41. Two point charges $2\sqrt{2}\mu\text{C}$ and $5\mu\text{C}$ are kept at point $(1,2,3)\text{m}$ and $(4,6,8)\text{m}$ respectively. coulomb force on $5\mu\text{C}$ charge :-

(1) $36 \times 10^{-5} (3\hat{i} + 4\hat{j} + 5\hat{k}) \text{ N}$
(2) $2 \times 10^{-5} (3\hat{i} + 4\hat{j} + 5\hat{k}) \text{ N}$
(3) $1 \times 10^{-5} (3\hat{i} + 4\hat{j} + 5\hat{k}) \text{ N}$
(4) $4 \times 10^{-5} (3\hat{i} + 4\hat{j} + 5\hat{k}) \text{ N}$

42. Find net force on charge Q placed on X-axis.

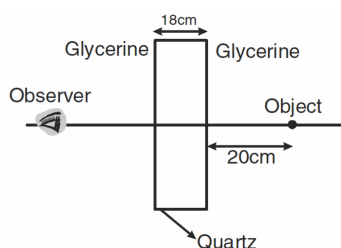


- (1) $\frac{\sqrt{2}Q^2}{8\pi\epsilon_0 R^2}$ (2) $\frac{\sqrt{2}Q^2}{4\pi\epsilon_0 R^2}$
 (3) $\frac{\sqrt{2}Q^2}{8\pi\epsilon_0 R}$ (4) $\frac{2\sqrt{2}Q^2}{8\pi\epsilon_0 R^2}$

43. A point source of light is placed at bottom of a water lake. If the circular illuminated area on the surface is equal to 4 times the square of depth of the lake, the refractive index of the water is

- (1) $\frac{\sqrt{\pi+2}}{2}$ (2) $\frac{\sqrt{\pi+4}}{2}$
 (3) $\sqrt{\pi+1}$ (4) $\frac{\sqrt{\pi+2}}{4}$

44. Given that velocity of light in quartz = 1.5×10^8 m/s and velocity of light in glycerine = $(9/4) \times 10^8$ m/s. Now a slab made of quartz is placed in glycerine as shown. The shift of the object produced by slab is :



- (1) 6 cm (2) 3.55 cm
 (3) 9 cm (4) 2 cm
45. A dentist wants a small mirror that when placed 2 cm from a tooth, will produce an large erect image three times the size. What kind of mirror must be used and what must its focal length be:

- (1) Concave mirror, 3.0 cm
 (2) Concave mirror, 1.5 cm
 (3) Convex mirror, 3.0 cm
 (4) Convex mirror, 1.5 cm

46. A driver stops his car at a red light. The car is fitted with side view mirror of focal length 10 m. An ambulance is approaching the car at a constant speed of 16 ms^{-1} . The speed of image of the ambulance, as seen by the driver in the side view mirror, when it is at distance of 50 m from the mirror will be

- (1) 1 ms^{-1} (2) $\frac{4}{9} \text{ ms}^{-1}$
 (3) $\frac{2}{3} \text{ ms}^{-1}$ (4) 2 ms^{-1}

47. The magnification of an object is m_1 when placed 25cm from a mirror & m_2 when object is moved 15cm further away from its initial position. If $\frac{m_1}{m_2} = 4$, the focal length of the mirror is :

- (1) 10cm (2) 30cm (3) 15cm (4) 20cm

48. The distance between an object and its doubly magnified image by a concave mirror is:
 [Assume f = focal length]

- (1) $3f/2$
 (2) $2f/3$
 (3) $3f$
 (4) depends on whether the image is real or virtual.

49. Two full turns of the circular scale of screw gauge cover a distance of 1 mm on scale. The total number of divisions on circular scale is 50. Further, it is found that screw gauge has a zero error of -0.03 mm . While measuring the diameter of a thin wire a student notes the main scale reading of 3 mm and the number of circular scale division in line, with the main scale as 35. The diameter of the wire is

- (1) 3.32 mm (2) 3.73 mm
 (3) 3.67 mm (4) 3.38 mm

50. One centimetre on the main scale of vernier callipers is divided into ten equal parts. If 10 divisions of vernier scale coincide with 8 small divisions of main scale, then least count of vernier callipers is :-

- (1) 0.01 cm (2) 0.02 cm
 (3) 0.05 cm (4) 0.005 cm

SECTION-A (CHEMISTRY)

51. For the reaction $A \rightarrow B$, the rate law expression is :Rate = $k[A]$. Which of the following statements is **incorrect** ?

- (1) The reaction is said to follow first-order kinetics
- (2) The half-life of the reaction will depend on the initial concentration of the reactant
- (3) K is constant for the reaction at a constant temperature
- (4) The rate law provide a simple way of predicting the rate at given concentration of reactants

52. A substance 'A' decomposes by a first-order reaction starting initially with $[A] = 2.00$ M and after 10 min $[A] = 0.5$ M. For this reaction what is the value of rate constant (K) ?

- (1) $1.38 \times 10^{-1} \text{ min}^{-1}$
- (2) $0.69 \times 10^{-1} \text{ min}^{-1}$
- (3) $3.29 \times 10^{-2} \text{ min}^{-1}$
- (4) $4.40 \times 10^{-2} \text{ min}^{-1}$

53. Initial amount of a reactant in first order reaction is 16 g. Half-life of reaction is 2 min. What amount of reactant will remain after 8 min ?

- (1) 2g (2) 1g (3) 3g (4) 4g

54. Rate constant for a zero order reaction is $2 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$.

If the concentration of the reactant after 25 s is 0.5 M, the initial concentration must have been.

- (1) 0.5 M
- (2) 1.25 M
- (3) 12.5 M
- (4) 1.0 M

55. The reaction,
 $3\text{I}^{-}(\text{aq}) + \text{S}_2\text{O}_8^{2-}(\text{aq}) \rightarrow \text{I}_3^{-}(\text{aq}) + 2\text{SO}_4^{2-}(\text{aq})$
 yields the kinetic data in the table.

$[\text{I}^{-}]_0$ (mol L ⁻¹)	$[\text{S}_2\text{O}_8^{2-}]_0$ (mol L ⁻¹)	Relative Rate
0.001	0.001	1
0.002	0.001	2
0.002	0.002	4

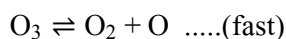
What is the rate equation ?

- (1) Rate = $k [\text{I}^{-}] [\text{S}_2\text{O}_8^{2-}]$
- (2) Rate = $k [\text{I}^{-}]^2 [\text{S}_2\text{O}_8^{2-}]$
- (3) Rate = $k [\text{I}^{-}]^3 [\text{S}_2\text{O}_8^{2-}]$
- (4) Rate = $k [\text{I}^{-}]^2 [\text{S}_2\text{O}_8^{2-}]^2$

56. 75% of a 1st order reaction gets completed in 4 hours. Calculate the time taken for 99.9% reaction to be completed :-

- (1) 2 hours
- (2) 20 hours
- (3) 200 hours
- (4) .2 hours

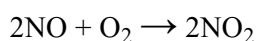
57. The chemical reaction $2\text{O}_3 \rightarrow 3\text{O}_2$ proceed as follows



The rate law expression should be

- (1) $R = k[\text{O}_3]^2$
- (2) $R = k[\text{O}_3]^2 [\text{O}_2]^{-1}$
- (3) $R = k[\text{O}_3]^1 [\text{O}_2]^1$
- (4) Unpredictable

58. What is the molecularity of following elementary reaction?



- (1) 2 (2) 1 (3) 4 (4) 3

59. In a reaction $2A \rightarrow \text{Products}$, the concentration of A decreases from 0.8 mol L^{-1} to 0.3 mol L^{-1} in 5 minutes then what is the average rate during this interval ?

- (1) $2.5 \times 10^{-2} \text{ M. min}^{-1}$
- (2) $10^{-1} \text{ M. min}^{-1}$
- (3) $1.25 \times 10^{-2} \text{ M. min}^{-1}$
- (4) $5 \times 10^{-2} \text{ M min}^{-1}$

60. **Statement-I** : The rate of a reaction decreases with passage of time as the concentration of reactant decrease.

Statement-II : Value of rate constant decreases with decreases in concentration of reactant.

- (1) Statement-I & II both are true
- (2) Statement-I is false and Statement-II is true.
- (3) Statement-I is true and Statement-II is false.
- (4) Statement-I & II both are false

61. The factors affecting the rate constant of reaction are :-

- (I) Temperature
- (II) Pressure
- (III) Concentration of reactant or product
- (IV) Catalyst

Choose the correct option from the alternative given below :-

- (1) I, II and IV
- (2) I, III and IV
- (3) I & IV
- (4) All of these

62. Match the terms of column I with column II

	Column-I		Column-II
A.	Inversion of cane sugar	p.	Zero order reaction
B.	Radioactive decay	q.	Pseudo first order reaction
C.	Thermal decomposition of HI on gold surface	r.	First order reaction

- (1) $A \rightarrow p, B \rightarrow q, C \rightarrow r$
- (2) $A \rightarrow q, B \rightarrow r, C \rightarrow p$
- (3) $A \rightarrow r, B \rightarrow q, C \rightarrow p$
- (4) $A \rightarrow q, B \rightarrow p, C \rightarrow r$

63. The decomposition of ammonia on platinum surface is zero order reaction. What is the rate production of H_2 if $k = 5 \times 10^{-4} \text{ mol}^{-1} \cdot \text{L} \cdot \text{s}^{-1}$?

- (1) $3 \times 10^{-4} \text{ mol. L. s}^{-1}$
- (2) $5 \times 10^{-4} \text{ mol. L. s}^{-1}$
- (3) $1.5 \times 10^{-3} \text{ mol}^{-1} \cdot \text{L} \cdot \text{s}^{-1}$
- (4) $1.5 \times 10^{-3} \text{ mol L}^{-1} \text{s}^{-1}$

64. In the reaction $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$, the rate of decomposition of H_2O_2 is 0.68 kg hr^{-1} . Find rate of formation of O_2 ?

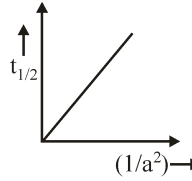
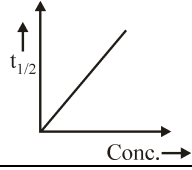
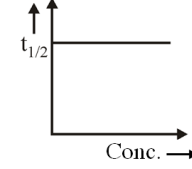
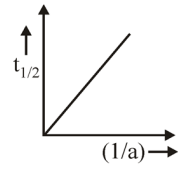
- (1) 0.34 kg hr^{-1}
- (2) 0.68 kg hr^{-1}
- (3) 0.02 kg hr^{-1}
- (4) 0.32 kg hr^{-1}

65. $A + 2B + C \rightarrow \text{Product}$
(Excess)

If concentration of (C) becomes double then what will be the effect on rate of reaction while concentration (A) and (B) remain constant ?

- (1) Become double
- (2) Remain same
- (3) Becomes half
- (4) Becomes four times

66. Match the column

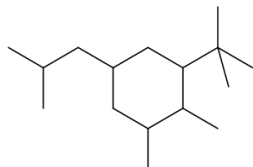
	Column-I		Column-II
(A)	Zero order	(P)	
(B)	First order	(Q)	
(C)	Second order	(R)	
(D)	Third order	(S)	

- (1) A-Q, B-R, C-S, D-P
- (2) A-R, B-Q, C-S, D-P
- (3) A-Q, B-R, C-P, D-S
- (4) A-P, B-Q, C-R, D-S

67. What is the order of a reaction which has a rate expression $r = k[A]^{1/2}[B]^{-1}[C]^{3/2}$

- (1) Two (2) One (3) Three (4) Zero

68. Number of 1° , 2° , 3° and 4° C present in compound are respectively



- (1) 7, 4, 5 and 1 (2) 7, 3, 4 and 1
(3) 7, 3, 5 and 1 (4) 7, 4, 4 and 1

69. The IUPAC name of is:

- (1) 3-Methylpent-1-en-4-al
(2) 3-Methylpent-4-enal
(3) 3-Methylpent-4-carbaldehyde
(4) 3-Methyl-5-oxopent-1-ene

70. **Assertion** :- Isopropyl alcohol is a secondary alcohol.

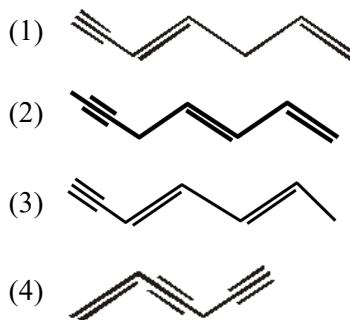
Reason :- —OH attached with 2° carbon.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) If Assertion is True but the Reason is False.
(4) If both Assertion & Reason are False.

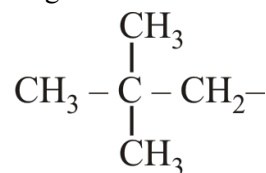
71. IUPAC name of is :-

- (1) 2-Hydroxy pentan-4-one
(2) 4-Hydroxy pentan-2-one
(3) 2-Oxo pentan-4-ol
(4) 4-Hydroxy-2-oxo pentane

72. Which of the following is structure of hepta-3,6-dien-1-yne ?

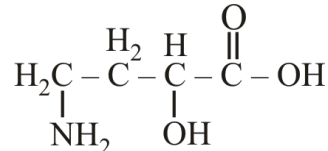


73. IUPAC name of given radical is :-



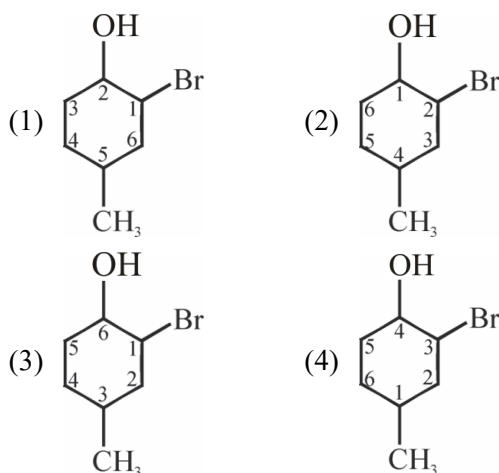
- (1) 2,2-dimethylbutyl
(2) 2,2-dimethyl propyl
(3) 2-methyl-2-methylene propane
(4) Pentyl

74. IUPAC name of the compound :

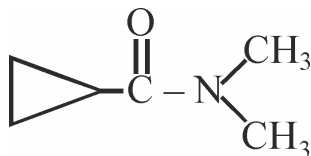


- (1) 2-amino-4-hydroxybutyric acid
(2) 4-amino-2-hydroxybutanoic acid
(3) 4-amino-2-hydroxybutyric acid
(4) 1-amino-3-hydroxybutanoic acid

75. Which numbering is right for IUPAC naming.

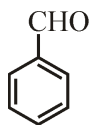


76. IUPAC name of the following compound is



- (1) N, N-Dimethyl cyclopropanamide
 (2) N-Methyl cyclopropanamide
 (3) N, N-Dimethyl cyclopropane carboxamide
 (4) N, N-Dimethyl cyclopropalonyl amide

77. Number of σ and π bonds in given compound is



- (1) 12, 4 (2) 14, 4
 (3) 12, 3 (4) 14, 3

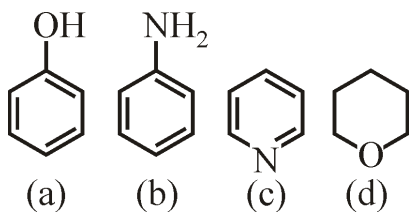
78. Write general formula of amide having 1 olefinic bond and 1 acetylenic bond.

- (1) $C_nH_{2n-5}NO$ (2) $C_nH_{2n-4}NO$
 (3) $C_nH_{2n+2}NO$ (4) $C_nH_{2n-6}NO$

79. Select incorrect match

	Molecule	First member
(1)	Alkene	$CH_2=CH_2$
(2)	Aldehyde	CH_3-CHO
(3)	Alcohol	CH_3-OH
(4)	Carboxylic acid	$H-COOH$

80. Which of the following are homocyclic compound ?

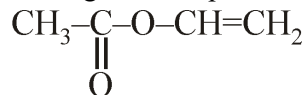


- (1) a, b, c, d
 (2) a, b, c
 (3) a, b
 (4) a, c

81. In which of the following compound electron pair is present in sp^3 orbital.

- (1) $CH_3-C\equiv C^-$ (2) $CH_2=\overset{\ominus}{C}H$
 (3) $CH_3-\overset{\ominus}{C}H_2$ (4) None

82. Common name of given compound is



- (1) Ethyl acetate
 (2) Vinyl acetate
 (3) Vinyl formate
 (4) Ethyl acetate

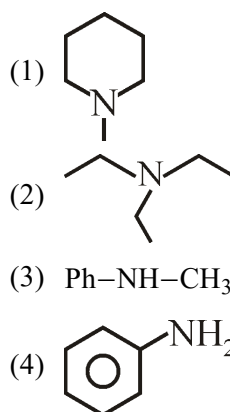
83. Select incorrect option :-

- (1) $CH_3-CH<$ Ethylene
 (2) $CH_3-CH(CH_3)-CH_2-$ Isobutyl
 (3) $CH_3-CH(CH_3)-$ Isopropyl
 (4) $CH_2=CH-$ Vinyl

84. What is prefix for amide ?

- (1) Carbamoyl
 (2) Carboxamide
 (3) Amide
 (4) Amino

85. Identify 2° amine :-



SECTION-B (CHEMISTRY)

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86. For the chemical reaction $-\frac{d[A]}{dt} = k[A]^2$, the unit for rate constant is :-
- time
 - minute⁻¹
 - s⁻¹ mol⁻¹ litre⁺¹
 - minute⁻¹ mol⁻¹ litre⁻¹
87. What is the activation energy for the reverse of this reaction ?
 $N_2O_4(g) \rightarrow 2NO_2(g)$
 Data for the given reaction :
 $\Delta H = +54 \text{ kJ}$ and $E_a = +57.2 \text{ kJ}$
- 54 kJ
 - +3.2 kJ
 - +60.2 kJ
 - +111.2 kJ
88. In a reaction the half life becomes one fourth on doubling the initial concentration of reactant then order of reaction is :-
- 0
 - 1
 - 2
 - 3
89. For an endothermic reaction, ΔH represents the enthalpy of the reaction in kJ/mol, the minimum energy of activation will be :-
- Less than ΔH
 - Zero
 - More than ΔH
 - Equal to ΔH
90. If fraction of active molecules in a chemical reaction are 10^{-5} , then calculate activation energy of this reaction at 27°C ?
- 28.72 kcal
 - 2.87 kcal
 - 6909 kcal
 - 6.909 kcal

91. A gaseous reaction, $A(g) \rightarrow B(g) + \frac{1}{2}C(g)$, Shows increase in pressure from 150 mm to 200 mm in 10 min. What is the rate of disappearance of A(g)?
- 20 mm.min⁻¹
 - 10 mm.min⁻¹
 - 5 mm.min⁻¹
 - 8 mm.min⁻¹
92. The rate of reaction increases by the increase of temperature because :-
- Collision frequency in increased
 - Energy of product decreases
 - Fraction of molecules possessing energy $\geq E_T$ (threshold energy) increases.
 - Mechanism of reaction is changed
93. **Statement-I** : Decomposition of gaseous ammonia on a hot platinum surface is a zero order reaction at high pressure.
Statement-II : At high pressure, the metal surface gets saturated with gas molecules.
- Both statement-I and statement-II are false
 - Statement-I is true but statement-II is false
 - Statement-I is false but statement-II is true
 - Both statement-I and statement-II are true
94. IUPAC name of given compound is
- $$\begin{array}{c}
 \text{CH}_3 \quad \text{CH}_2 \\
 | \quad \parallel \\
 \text{HO} - \text{C} - \text{CH} \\
 | \\
 \text{CH} \\
 / \quad \backslash \\
 \text{H}_3\text{C} \quad \text{CH}_3
 \end{array}$$
- 2-Isopropyl pent-1-ene-2-ol
 - 3,4-dimethyl pent-3-en-3-ol
 - 3,4-dimethyl pent-1-ene-3-ol
 - None of these

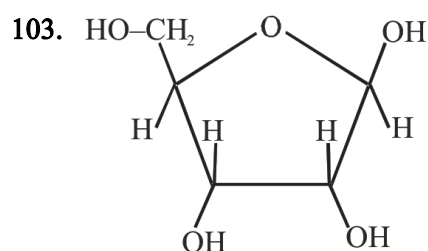
SECTION-A (BIOLOGY-I)

101. Which of the following compounds are pyrimidines ?

- (1) Adenine, Guanine, Thymine
- (2) Thymine, Cytosine, Uracil
- (3) Adenine, Uracil, Guanine
- (4) Thymine, Uracil, Adenine

102. Adenine + Ribose + Phosphate form :-

- (1) Adenosine
- (2) Deoxyadenylic acid
- (3) Adenylic acid
- (4) Adenine



- (1) It is structure of deoxyribose
- (2) It is component of deoxyribonucleotide
- (3) It is non reducing monosaccharide
- (4) It is present in RNA

104. Which of the following feature provide stability to helical structure of DNA ?

- (1) Presence of hydrogen bonds between two strands
- (2) The plane of one base pair stacks over the other in double helix
- (3) Parallel polarity of two strands
- (4) Both (1) and (2)

105. In DNA molecule, at each step of ascent. The strand turns :-

- (1) 120° (2) 360° (3) 30° (4) 36°

106. According to double helix model, distance between two polynucleotides chains in DNA remains almost constant because.

- (1) In most of cases DNA functions as a genetic material.
- (2) Always a purine comes opposite to a pyrimidine
- (3) Histones are involved in packaging of DNA Helix
- (4) All of these

107. Replication does not initiate randomly at any place in DNA. There is a definite region where the replication originate such regions are called

- (1) Replication fork
- (2) Origin of replication
- (3) Promoter site
- (4) Palindromic Sequence

108. Hershey and Chase used which organism to prove that DNA is genetic material ?

- (1) Streptococcus (2) TMV
- (3) T₂ - bacteriophage (4) λ - bacteriophage

109. During replication of DNA, deoxynucleotides are added one by one in 5' to 3' direction by phosphodiester bond formation. In this process of polymerisation, the source of energy is :-

- (1) ATPs (2) GTPs
- (3) dNTPs (4) NADH₂

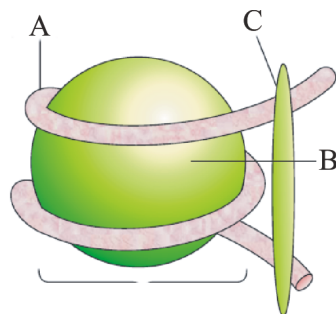
110. RNA can easily express the character because :-

- (1) RNA can directly code for the synthesis of protein
- (2) RNA mutate at faster rate
- (3) RNA is dependent on DNA for protein synthesis
- (4) RNA is unstable

111. Main enzyme of DNA replication is :-

- (1) DNA dependent RNA polymerase
- (2) DNA dependent DNA polymerase
- (3) RNA dependent RNA polymerase
- (4) RNA dependent DNA polymerases

112. Identify A,B and C in given diagram :-



	A	B	C
(1)	Histone H ₁	DNA	Histone octamer
(2)	DNA	Histone octamer	Histone H ₁
(3)	Histone octamer	Histone H ₁	DNA
(4)	Histone octamer	DNA	Histone H ₁

113. Which structure in chromatin are seen as "beads-on-string" structure when viewed under electron microscope ?

- (1) Nucleoid
- (2) Nucleosomes
- (3) Nucleolus
- (4) Histone

114. The packaging of chromatin in eukaryotes at higher level requires set of proteins that are collectively referred to as :-

- (1) Histones
- (2) Polyamines
- (3) Non-Histone Chromosomal protein
- (4) Globins

115. Heterocyclic ring is found in :-

- (1) Nitrogen bases
- (2) Pentose sugar
- (3) Lecithin
- (4) All of the above

116. If the DNA of virus is labelled with ³²P and the protein coat of virus is labelled with ³⁵S after transduction which molecule would be present inside bacterial cell?

- (1) ³²P only
- (2) ³⁵S only
- (3) Both ³⁵S and ³²P
- (4) Neither molecule would be present inside the cell.

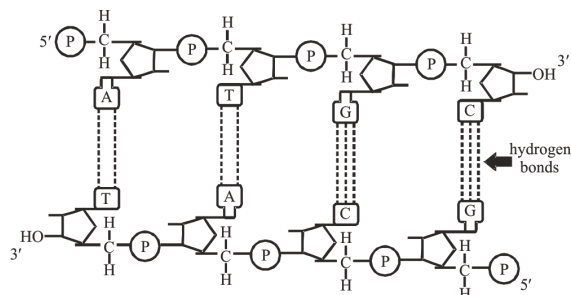
117. Find the correct match :-

- (1) $\phi \times 174$ phage - 5386 ribose sugars in genetic material
- (2) λ phage - 48502 ribonucleotides
- (3) *E. coli* - 4.6×10^6 bp
- (4) Human - 3.3×10^9 nucleotides in genome

118. If Meselson and Stahl's experiment is continued for three generations in bacteria the ratio of $^{15}\text{N}/^{15}\text{N} : ^{15}\text{N}/^{14}\text{N} : ^{14}\text{N}/^{14}\text{N}$ containing DNA in third generation would be :-

- (1) 0 : 1 : 7
- (2) 1 : 4 : 0
- (3) 0 : 1 : 3
- (4) 0 : 1 : 4

119. Study the diagram and identify the correct statement with reference to given polynucleotide chain :-



- (A) Sequence of bases on one strand cannot be predicted by sequence of other strand.
 (B) The two chain have parallel polarity. It means one chain has polarity $5' \rightarrow 3'$ the other has $3' \rightarrow 5'$ polarity.
 (C) There is uniform distance between two strand because a purine always pair with pyrimidine.
 (D) If each strand of it act as template for synthesis of new strand the two double stranded daughter cell DNA would be produced identical to parental.
- (1) A and B
 (2) B and C
 (3) C and D
 (4) A and D

120. Select the correct option:-

	Direction of DNA synthesis	Direction of reading of template DNA strand
(1)	$5' \rightarrow 3'$	$3' \rightarrow 5'$
(2)	$3' \rightarrow 5'$	$5' \rightarrow 3'$
(3)	$5' \rightarrow 3'$	$5' \rightarrow 3'$
(4)	$3' \rightarrow 5'$	$3' \rightarrow 5'$

121. Total percentage of A and T in DNA is 45%. The percentage of 'G' will be :-
- (1) 22.5% (2) 27.5%
 (3) 45% (4) 55%

122. The carbon atoms of the pentose sugar involved in phosphodiester bond formation in DNA and RNA are :-

- (1) C'_1 and C'_5 (2) C'_2 and C'_3
 (3) C'_3 and C'_5 (4) C'_4 and C'_5

123. Adenosine, guanosine, uridine, and cytidine are all _____ but adenylic acid, guanylic acid, uridylic acid and cytidylic acid are _____ :-

- (1) nucleotides, nucleosides
 (2) nucleosides, nucleotides
 (3) nucleotides, nucleic acids
 (4) nucleotides, nuclease

124. Choose the correct statement :-

- (1) Haploid content of human DNA is 4.6×10^6 bp.
 (2) A nitrogenous base is linked to pentose sugar through phosphodiester linkage.
 (3) X-ray diffraction data of Maurice Wilkins and Rosalind Franklin was the basis of Watson and Cricks DNA model.
 (4) DNA as an acidic substance was first identified by Watson and Crick.

125. Okazaki fragments are joined by :-

- (1) DNA ligase (2) DNA polymerase
 (3) Helicase (4) Primase

126. **Assertion** : The distance between two polynucleotide chains in dsDNA remains constant.

Reason : Always a purine comes opposite to pyrimidine or vice versa.

- (1) Both A and R are true and R is the correct explanation of A.
 (2) Both A and R are true but R is NOT the correct explanation of A.
 (3) A is true but R is false
 (4) A is false but R is true

127. Watson and crick proposed certain features for double helical model of structure of DNA. Which of the following is not a feature of double helical structure of DNA ?

- (1) Two chains are anti-parallel to each other.
- (2) Nitrogen bases of two strands are complementary and paired through covalent bonds.
- (3) Each DNA strand is made up of nucleotides.
- (4) Nitrogen bases are present towards inner side of double helix.

128. Choose the wrong statement regarding B-DNA ?

- (1) Right handed with 3.4 Å rise per base pair
- (2) Left handed with 10 base pairs per turn
- (3) Right handed with 10 base pairs per turn
- (4) Right handed with antiparallel nature of strands

129. In a nucleotide, an ester bond is present between

- (1) Carboxyl and phosphate group of sugar
- (2) Carbonate and carboxyl group of sugar
- (3) Phosphate and hydroxyl group of sugar
- (4) Sulphate and carboxyl group of sugar

130. **Assertion:** Presence of thymine at the place of uracil confers stability to DNA.

Reason: RNA is best for storage of genetic information.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

131. Which of the following DNA melt at lowest temperature.

- (1) 5' AATAAAG 3'
3' TTATTTTC 5'
- (2) 5' AATGCTG 3'
3' TTACGAC 5'
- (3) 5' ATGCTGC 3'
3' TACGACG 5'
- (4) 5' AGGCAGC 3'
3' TCCGTCG 5'

132. Which of the following statements about Griffith's experiment are correct?

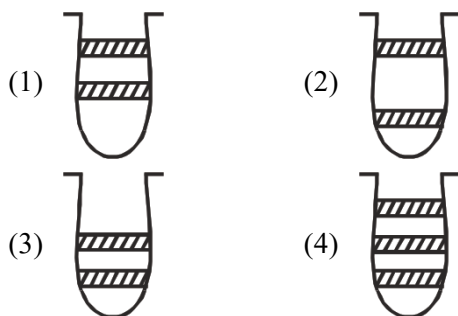
- I. S-strain have mucus (polysaccharide) coat.
- II. S-strain are virulent and cause pneumonia infection, while R-strain do not.
- III. Transforming principle is associated with genetic material of R-strain.
- IV. Transformation of R-strain into S-strain can take place in a test tube.

- (1) I and III
- (2) III and IV
- (3) I, II and IV
- (4) II, III and IV

133. First genetic material was (A) that is (B) and (C) as compare with best genetic material that is (D) :-

	A	B	C	D
(1)	DNA	Less reactive	more stable	RNA
(2)	RNA	More reactive	less stable	DNA
(3)	RNA	More reactive	more stable	DNA
(4)	DNA	Less reactive	less stable	RNA

134. Light DNA molecule is replicated in a heavy medium. After the four successive replication of DNA, 16 molecules of DNA are formed. Which one of the following is true for these DNA ?



135. How many enzymes/proteins are related to DNA replication process :

RNA ligase, RNA polymerase, Release factor, Helicase, DNA polymerase, DNA ligase, Polynucleotide phosphorylase

- (1) Eight (2) Four
(3) Seven (4) Five

SECTION-B (BIOLOGY-I)

136. Cytosine is :-

- (1) N-base (2) Nucleoside
(3) Nucleotide (4) Amino acid

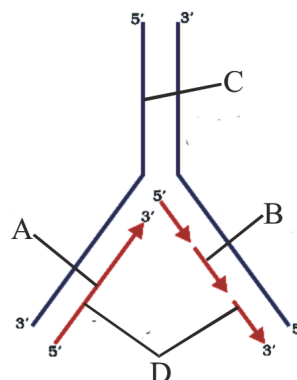
137. Pitch of B-DNA is :-

- (1) 3.4 nm
(2) 34 nm
(3) 34 cm
(4) 0.34 nm

138. In Oswald Avery, Colin MacLeod and Maclyn McCarty experiment transforming substance was digested by using :-

- (1) DNase
(2) RNase
(3) Proteases
(4) High temperature

139. Identify part A, B, C, D in the figure given below :-



	A	B	C	D
(1)	Continuous synthesis	Discontinuous synthesis	Template strand	Newly synthesised strands
(2)	Discontinuous synthesis	Continuous synthesis	Newly synthesised strand	Template strand
(3)	Template strand	Newly synthesised strand	Continuous synthesis	Discontinuous synthesis
(4)	Newly synthesised strand	Template strand	Continuous synthesis	Discontinuous synthesis

140. In which type of virus, rate of evolution is faster as compare to others :-

- (1) DNA genome and longer life span
(2) RNA genome and longer life span
(3) DNA genome and shorter life span
(4) RNA genome and shorter life span

141. In how many cases Normal transformation will occur :-

- A- Heat killed 'S' → injected in Mice
B- Heat killed 'R' → injected in Mice
C- Heat killed 'S' + R → injected in Mice
D- Heat killed 'R' + S → injected in Mice
E- Heat killed 'S' + R + Protease → injected in Mice
F- Heat killed 'S' + R + DNase → injected in Mice
G- Heat killed 'S' + R + RNase → injected in Mice
H- Heat killed 'S' + Heat killed 'R' → injected in Mice

- (1) 5 (2) 4 (3) 3 (4) 2

142. Which of the following is not a nucleotide ?

- (1) Adenylic acid
- (2) Uridylic acid
- (3) Deoxythymidylic acid
- (4) Deoxycytidine

143. In each strand of a double-stranded DNA the nucleotides are joined together by :

- (1) Phosphodiester bond
- (2) Hydrogen bond
- (3) Peptide bond
- (4) Disulphide bond

144. A molecule that can act as a genetic material must fulfill the traits given below, except

- (1) it should be able to express itself in the form of 'Mendelian characters'
- (2) it should be able to generate its replica
- (3) it should be unstable structurally and chemically
- (4) it should provide the scope for slow changes that are required for evolution

145. Read the following four statement (A-D) :-

- (A) Backbone of DNA is formed by sugar and phosphate.
- (B) DNA and RNA have similar pyrimidine bases.
- (C) There are three hydrogen bonds between G and C in DNA molecule.
- (D) At each step of ascent, strand of DNA turns around 36° .

How many of the above statements are correct ?

- (1) Four (2) Three (3) Two (4) One

146. Find out the correct statement about DNA.

- (1) It is a double helical structure made up of two polypeptide chains.
- (2) It is an acidic substance present in nucleus and was first identified by F. Miescher in 1869.
- (3) Phosphate group is linked to 5'-OH of the nucleoside by glycosidic linkage.
- (4) All four deoxyribonucleotides are always equally present in both the strands.

147. **Assertion** :- Adenosine is nucleoside.

Reason :- Adenine is purine N-base.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

148. **Assertion**: DNA is an acidic substance present in nucleus.

Reason: DNA has phosphate group which is responsible for acidic nature and it is present in chromosome in nucleus.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

149. The ratio of $\frac{A+T}{G+C}$ in one strand of dsDNA is 0.8. The similar ratio in complementary strand is :

- (1) 1.25 (2) 0.8 (3) 1 (4) 0.2

150. Assuming that 50 heavy (i.e. containing N^{15}) DNA molecules replicated twice in a medium containing N^{14} , we expect :-

- (1) 100 half heavy and half light and 150 light DNA molecules
- (2) 100 half heavy and half light and 100 light DNA molecules
- (3) 50 heavy and 150 light DNA molecules
- (4) 50 heavy and 100 light DNA molecules

SECTION-A (BIOLOGY-II)

151. How many of the following structures are present only in male?

Vas deferens, Ejaculatory duct, Urethra, Seminiferous tubule, Epididymis, Bartholin's gland, Corpus luteum, Tubule recti.

- (1) Six (2) Four
(3) Five (4) Seven

152. Find the incorrect match w.r.t. increase in the levels of following hormones :-

- (1) Oxytocin - Milk ejection
(2) Prolactin - Milk secretion
(3) Progesterone - Uterine contraction
(4) Luteinising hormone - Stimulates ovulation

153. Withdrawal of which hormone is the immediate cause of menstruation ?

- (1) Estrogen
(2) FSH
(3) FSH-LH
(4) Progesterone

154. Spermatocytogenesis is formation of :-

- (1) Spermatozoa from spermatids
(2) Secondary oocyte from primary oocyte
(3) Spermatids from spermatogonia
(4) Spermatogonia from spermatids

155. How many sperm and ova will be formed from 100 secondary oocytes and 100 secondary spermatocytes in human :-

- (1) 50 ova & 200 sperm
(2) 50 ova & 100 sperm
(3) 100 ova & 200 sperm
(4) 100 ova & 400 sperm

156. The number of mammary lobes in a breast is :

- (1) 10 - 20
(2) 15 - 20
(3) 20 - 25
(4) 30 - 40

157. Arrange the events in human female reproductive cycle

a—Secretion of FSH

b—Growth of corpus luteum,

c—Growth of follicle and oogenesis

d—Ovulation,

e—Sudden increase in level of LH

- (1) a—c—e—d—b
(2) c—a—d—b—e
(3) a—d—c—e—b
(4) b—a—c—d—e

158. Identify the correct match from the columns I, II and III.

	Column - I		Column - II		Column - III
(A)	Corpus luteum	(a)	Fallopian tube	(i)	Estrogen
(B)	Granulosa cells	(b)	Sperm	(ii)	Collection of ovum
(C)	Proximal centriole	(c)	Ovary	(iii)	Promote the cleavage
(D)	Fimbriae	(d)	Follicle	(iv)	Progesterone

- (1) A → d → i; B → c → iv,
C → b → iii, D → a → ii
(2) A → c → i; B → d → iii,
C → d → ii, D → b → iv
(3) A → c → iv; B → d → i,
C → b → iii, D → a → ii
(4) A → b → iv; B → c → iii,
C → a → ii, D → d → i

159. Assertion : All copulation cannot lead to fertilization and pregnancy.

Reason : Fertilization can only occur if the ovum and sperms are transported simultaneously to the ampullary isthmic junction.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

160. Which of the following hormone responsible for concentration of testosterone inside the seminiferous tubules ?

- (1) ICSH
- (2) ABP
- (3) Inhibin
- (4) Androgen

161. Given below are two statements :

Statement I : The release of sperms into the seminiferous tubules is called spermiation.

Statement II : Spermatogenesis is the process of formation of sperms from spermatogonia.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) Both **Statement I** and **Statement II** are incorrect
- (2) **Statement I** is correct but **Statement II** is incorrect
- (3) **Statement I** is incorrect but **Statement II** is correct
- (4) Both **Statement I** and **Statement II** are correct

162. Given below are two statements :

Statement-I : Primary oocyte grows in size and completes its first meiotic division within the secondary follicle.

Statement-II : During oogenesis first meiotic division are unequal division resulting in the formation of secondary oocyte and first polar body.

- (1) Both statement I and statement - II are incorrect.
- (2) Statement - I is correct but statement - II is incorrect.
- (3) Statement - I is incorrect but statement - II is correct.
- (4) Both statement - I and statement - II are correct.

163. Given below are two statements

Statement-I : Oogenesis is initiated during the embryonic development stage.

Statement-II : The secondary follicle is characterised by a fluid filled cavity called antrum.

In the light of above two statements, choose the most appropriate answer from the options given below.

- (1) Both statement-I and statement-II are correct
- (2) Both statement-I and statement-II are incorrect
- (3) Statement-I is correct but statement-II is incorrect
- (4) Statement-I is incorrect but statement-II is correct

164. Assertion (A) :- Colostrum from mother is essential for new born babies.

Reason (R) :- Colostrum is nutritious contains protein and antibodies.

- (1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**.
- (2) **(A)** is correct but **(R)** is not correct.
- (3) **(A)** is not correct but **(R)** is correct.
- (4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**.

165. Assertion :- Sperm formation continues even in old men but formation of ovum ceases in women around the age of fifty years.

Reason :- Human male having the longer life span than the female.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

166. Assertion (A):- The presence or absence of hymen is not a reliable indicator of virginity.

Reason (R):- In some women, the hymen persists even after coitus and it can also be broken by a sudden fall or jolt.

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (3) (A) is correct but (R) is not correct
- (4) (A) is not correct but (R) is correct

167. Menstrual cycle is initiated by :-

- (1) A sudden fall of FSH from anterior pituitary
- (2) A increase the level of oestrogen and decrease the level of progesteron
- (3) Low level of progesteron and oestrogen due to degeneration of corpus luteum
- (4) Low level of HCG due to degeneration of Graafian follicle

168. How many seminiferous tubules are present in one testicular lobule ?

- (1) Only one
- (2) 1 – 3
- (3) 10 – 20
- (4) 7 – 10

169. Correct sequence for milk ejection are :-

- (1) Mammary tubule - mammary duct - lactiferous duct - ampulla - nipple
- (2) Mammary duct - mammary tubule - ampulla - lactiferous duct - nipple
- (3) Mammary tubule - mammary duct - ampulla - lactiferous duct - nipple
- (4) Lactiferous duct - mammary tubule - mammary duct - ampulla - nipple

170. Which of the following statements about human sperms is correct.

- (1) Acrosomes serves no particular functions.
- (2) Acrosome has pointed end, penetrate egg membranes.
- (3) The sperm lysins in the acrosome dissolves the egg envelope facilitation fertilization.
- (4) Acrosome serves sensory structures leading the sperm towards ovum.

171. How many sperm a human male ejaculates in one ejaculation ?

- (1) 20-25 million sperms
- (2) 200-300 million sperms
- (3) 200-500 million sperms
- (4) 800-900 millions sperms

172. Which hormone is responsible for growth of alveoli in breast :-

- (1) Oestrogen
- (2) Progesterone
- (3) PRL
- (4) Oxytocin

173. Which of the following is a group of ovarian steroid hormones?

- (1) GnRH, FSH and LH
- (2) FSH, LH and Estrogen
- (3) Testosterone, Estrogen and Progesterone
- (4) Estrogen and Progesterone

174. Which one of the following structure of sperm is mismatched?

(1)	Acrosome	Contain sperm lysins that help in fertilisation
(2)	Nucleus	Contain haploid genome
(3)	Proximal centriole	Starts cleavage in zygote
(4)	Tail	Contain numerous mitochondria, which produce energy for movement of tail

175. The spermatids are transformed into spermatozoa by sertoli cells due to which hormone ?

- (1) LH (2) FSH
(3) Androgen (4) ABP

176. Which of the following option represents haploid and diploid stages of gametogenesis respectively ?

- (1) Primary spermatocytes and secondary spermatocytes
(2) Spermatogonia and primary spermatocytes
(3) Spermatogonia and sperms
(4) Secondary spermatocytes and primary spermatocytes.

177. During early pregnancy, which structure secrete H.C.G. for maintenance corpus luteum ?

- (1) Hypothalamus and pituitary
(2) Trophoblast
(3) Vagina
(4) Placenta

178. Which hormone is responsible for ovulation ?

- (1) GnRH (2) FSH
(3) LH (4) Progesterone

179. Which hormone attains two peaks during menstrual cycle ?

- (1) Inhibin (2) LH
(3) Progesterone (4) Estrogen

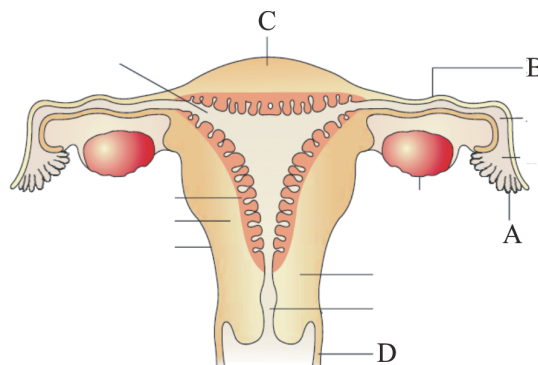
180. Identify the hormone, which do not attain peak before ovulation (during menstrual cycle) ?

- (1) FSH
(2) LH
(3) Estrogen
(4) Progesterone

181. Which layer of uterus degeneration leads to bleeding during menstrual cycle ?

- (1) Perimetrium
(2) Myometrium
(3) Stratum functionalis
(4) Stratum basalis

182. In the given diagram identify A, B, C, D :



	A	B	C	D
(1)	Fimbriae	Isthmus	Uterine fundus	Vagina
(2)	Fimbriae	Uterine fundus	Vagina	Isthmus
(3)	Vagina	Isthmus	Uterine fundus	Fimbriae
(4)	Fimbriae	Uterine fundus	Isthmus	Vagina

183. In a female undergone hysterectomy which one of the following event will not occur ?

- (1) Fertilisation
- (2) Menstruation
- (3) Follicular development
- (4) Both (1) & (2)

184. Find out correct sequence of menstrual cycle's phase?

- (1) Ovulation, Bleeding phase, Luteal phase
- (2) Bleeding phase, ovulation, Postovulatory phase, Progesteronic phase
- (3) Menstrual phase, Oestrogenic phase, Ovulation, Secretory phase
- (4) Bleeding phase, Ovulation, Oestrogenic phase

185. Fertilisation can only occur if the ovum and sperms are transported simultaneously to the Fill in the blank space with correct option:

- (1) Isthmus part of fallopian tube
- (2) Ampulla part of fallopian tube
- (3) Uterus
- (4) Vagina

SECTION-B (BIOLOGY-II)

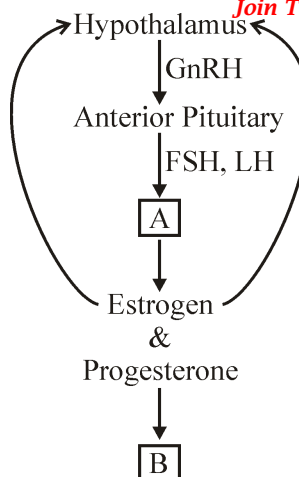
186. The part of Fallopian tube closest to the ovary is :

- (1) Isthmus
- (2) Infundibulum
- (3) Cervix
- (4) Ampulla

187. Which glands secretion help to destroy the acidic medium present in male urethra :-

- (1) Secretions of seminal vesicle
- (2) Secretions of cowpers gland
- (3) Secretions of prostate gland
- (4) Secretions of preputial gland

188.



Find out 'A' and 'B' respectively :-

- (1) Ovary and Uterus
- (2) Uterus and Ovary
- (3) Oviduct and Ovary
- (4) Seminal vesicle and Uterus

189. Which of the following is not a secondary sex organ:-

- (1) Epididymis
- (2) Prostate gland
- (3) Ejaculatory duct
- (4) Testis

190. Which external genitalia of female is homologous to male penis ?

- (1) Labia majora
- (2) Labia minora
- (3) Hymen
- (4) Clitoris

191. **Statement-I** : Spermatids are transformed into spermatozoa by the process called spermiogenesis.

Statement-II : Acrosome possesses numerous mitochondria which produces energy for the movement of tail.

- (1) Statement I and II both are correct
- (2) Statement I and II both are incorrect
- (3) Only Statement I is correct
- (4) Only Statement II is correct

192. **Statement-I** : The endometrium undergoes cyclic changes during menstrual-cycle.

Statement-II : Change in uterus are induced by change in level of ovarian hormones.

- (1) Both Statement-I and Statement-II are incorrect.
- (2) Both Statement-I and Statement-II are correct.
- (3) Statement-I is correct but Statement-II is incorrect.
- (4) Statement-I is incorrect but Statement-II is correct.

193. **Statement-I** : Myometrium exhibits strong uterine contraction during delivery of the baby.

Statement-II : Myometrium is the muscular layer present in the uterus.

- (1) Statement-I and II both are correct.
- (2) Statement-I and II both are incorrect
- (3) Only statement-I is correct.
- (4) Only statement-II is correct.

194. **Assertion (A)** : Ovaries are the primary sex organs of female.

Reason (R) : Ovaries produce gametes and no hormones.

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (3) (A) is correct but (R) is not correct
- (4) (A) is not correct but (R) is correct

195. **Assertion** :- Regular cyclic menstruation is an indicator of normal reproductive phase.

Reason :- It occurs in whole life of normal female.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

196. The liberation of sperms from body of male is called as:

- (1) Semination
- (2) Spermiation
- (3) Ejaculation
- (4) Insemination

197. Number of testicular lobules in both testes are :-

- (1) 250
- (2) 500
- (3) 750
- (4) 200-300

198. The thick vertical septum formed by tunica albuginea is known as :-

- (1) Rete testis
- (2) Mediastinum testis
- (3) Tubuli recti
- (4) Mesovarium

199. Which of the following part of sperm contains chromosomal material ?

- (1) Acrosome (2) Nucleus
(3) Middle piece (4) Tail

200. Which of the part of human sperm called as energy chamber of sperm ?

- (1) Head (2) Middle
(3) Neck (4) Tail



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